

Muhammad Najid

Electrical Engineer (Embedded Systems)

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OBJECTIVES

Embedded Systems Engineer with hands-on experience in **FPGA, STM32, ESP32, and FreeRTOS**, specializing in firmware development, hardware interfacing, and IoT systems. Strong focus on building reliable, real-time embedded solutions with efficient debugging and system optimization.

EDUCATION

National University of Computer and Emerging Sciences

BS Electrical Engineering (Computer System Engineering)

August 2022– 8th May, 2026

EXPERIENCE

Pakistan Civil Aviation Authority (PCAA) - Intern

Electrical Engineering Department

23 June 2025 – 19 July 2025

- Analyzed generator synchronization and ATS panel systems
- Assisted in inspection and maintenance of airport electrical infrastructure
- Studied switchgear protection systems (OCB, VCB, SF6) in real-world applications

TECHNICAL SKILLS

Microcontrollers: STM32, ESP32, AVR (ATmega328P), Raspberry Pi, Jetson Nano

Programming: C++, Assembly, Python (basic)

RTOS: FreeRTOS (tasks, queues, semaphores)

Protocols: UART, SPI, I2C, CAN

Hardware Design: PCB Design (KiCad/Proteus), Schematic Design

Tools: STM32CubeIDE, MPLAB X, Arduino IDE

IoT: WiFi, BLE, Adafruit IO, AWS (basic)

PROJECTS

Intelligent Fire Detection & Suppression System (Final Year Project)

- Developed AI-based fire detection system using YOLO for real-time monitoring
- Integrated embedded hardware for automated suppression and alerting
- Implemented IoT-based emergency notification system

Traffic Light Controller using FSM (Verilog HDL)

- Designed FSM-based traffic light controller with RED, YELLOW, and GREEN states
- Implemented synchronous state transitions using Verilog HDL
- Simulated and verified functionality using testbench in Xilinx Vivado

Air Quality Monitoring System (ATmega328P + MQ-135)

- Designed embedded system with LCD and keypad interface
- Implemented real-time data display and user navigation menu

IoT Sensor Data Monitoring (Adafruit IO)

- Published real-time sensor data using Python and cloud dashboards
- Integrated IR and PIR sensors with alert mechanisms

RELATED COURSEWORK

- Digital Logic Design
- Microprocessors & Microcontrollers
- Embedded Systems
- Computer Architecture
- Signals & Systems
- Electronic Circuit Design

CERTIFICATES

- Academic English: Writing Specialization – University of California, Irvine (Coursera) — *Completed: June 2025*
- Freelancing Training Program (DigiSkills.pk – Virtual University of Pakistan) — *Completed: October 2022*